Summary of Aydogan Ozcan's group's DNN Papers

Quick index (requested front matter).

1) Papers with experiments.

- **Paper numbers:** 1, 2, 5, 9, 10, 15, 16, 17, 19, 20, 22, 23, 24, 26, 27, 28, 30, 31, 32, 33, 34, 35.

2) Summary of mentioned D^2NN applications across the listed papers.

- **All-optical classification / recognition:** 1, 3, 4, 5, 6, 8, 9, 14, 16, 34.
- **Spectral processing and multispectral imaging:** 2, 9, 20.
- **Imaging, reconstruction, and lens-like transformations:** 1, 11, 14, 20, 24, 27, 31.
- **Quantitative phase imaging (QPI) and phase recovery:** 13, 21.
- **Pulse shaping / temporal waveform engineering:** 10.
- **Universal linear optical transforms and permutation operators:** 12, 18, 19, 25, 32.
- **Encryption and information hiding / concealment cameras:** 17, 25, 26.
- **Diffuser/occlusion-robust vision and communication:** 16, 21, 22, 34.
- **Hidden-object and defect sensing:** 23.
- **Phase conjugation / wavefront correction:** 28, 34.
- **All-optical denoising and preprocessing:** 30.
- **Nonlinear function computation/encoding:** 29, 33.
- **Structural health monitoring (vibration-spectrum estimation):** 35.

1. All-optical machine learning using diffractive deep neural networks (2018)

DOI. 10.1126/science.aat8084

Innovation. This is the foundational D^2NN paper. The key step was to treat a stack of passive diffractive surfaces as a trainable physical neural network: each point on each layer is optimized in software, then fabricated, and inference happens when light simply propagates through the stack. The paper is not only about classification; it also shows that the same training framework can

learn optical functions such as lensing/imaging. In other words, it established the *new paradigm* that "weights" do not need to live in electronics—they can be encoded into the geometry of passive optical layers, so the desired transformation is executed at the speed of light and with almost no inference power draw.

Experiment / system setup. Status: physical THz demonstration, and the accessible supplementary material is unusually informative. The classifier D^2NNs used a **5-layer transmissive 3D-printed architecture**. For the classification devices, the input aperture was 8×8 **cm**; the trained phase values were converted to a height map using $\Delta z = \lambda\Delta\phi/(2\pi\Delta n)$, and a uniform **0.5 mm substrate** was added to every printed layer. The layers were fabricated using **Objet30 Pro 3D** (**Stratasys**) with **VeroBlackPlus RGD875**. At **0.4 THz**, the reported material constants were $n = 1.7227$ and $k = 0.0311$, corresponding to an attenuation coefficient of about 520.7 m^{-1}. For the classifier systems, the axial spacing between successive layers was **3.0 cm**; for the imaging/lens D^2NN, it was much more compact at **4.0 mm**. The THz bench used a **WR2.2 amplifier/multiplier chain** (Virginia Diode Inc.) driven by a **16 dBm** RF tone at **11.111 GHz**, multiplied by 36× to generate continuous-wave **0.4 THz** illumination. The output plane was scanned with a **VDI mixer/AMC detector** mounted on **two Thorlabs NRT100** stages. The reported detector scan steps were about **1.2 mm** for the MNIST classifier, **1.6 mm** for the Fashion-MNIST classifier, and **600 μm** for the imaging network. The detector plane was placed **3 cm** after the last layer for the classifiers and **7 mm** after the last layer for the lens/imaging design. The supplement also reports beam spot sizes of about **40 mm**, **104 mm**, and **20 mm** FWHM at the object plane for the digit-classification, fashion-classification, and imaging experiments, respectively, and cites a printer resolution of **600 dpi** with about **0.1 mm** accuracy. So, after reading the supplement, this entry is no longer just "a 5-layer printed THz classifier" in general terms: it is a fairly well specified **five-layer, passive, transmissive THz D^2NN platform** with concrete fabrication and bench parameters.

2. Design of task-specific optical systems using broadband diffractive neural networks (2019)

DOI. 10.1038/s41377-019-0223-1

Innovation. This paper extends D^2NNs from monochromatic coherent classification to **broadband, task-specific spectral processing**. The conceptual jump is that the network is trained not merely to map one image to another, but to use wavelength as part of the computation. The examples in the paper include **spectral filtering** and **wavelength demultiplexing**, showing that a diffractive network can be optimized as a learned broadband optical component rather than a narrowband classifier. This is a *new method* for optical-system design: instead of hand-designing a filter bank or demultiplexer, the authors jointly optimize the diffractive surfaces for the end task over a spectrum.

Experiment / system setup. Status: physical broadband THz demonstration. The experimental system used a **3-layer transmissive diffractive network**. The accessible methods text reports a **temporally incoherent broadband THz source** generated in a THz time-domain spectroscopy setup using a **Ti:Sapphire Mira-HP laser at 780 nm**; detection electronics included a Femto DHPCA-100 amplifier and a Zurich Instruments MFLI lock-in. Fabrication details are unusually clear in the paper: the layers were printed using **Objet30 Pro** with **VeroBlackPlus RGD875**; the smallest printable lateral feature was **0.5 mm**; training used an oversampled grid of **0.125 mm**; each neuron had **16 thickness levels** with **62.5 μm** steps; the **base thickness was 0.5**

mm and the maximum trainable phase-modulation thickness was **1 mm**. The first layer had an active area of about 1×1 **cm**, the others about 5×5 **cm**.

3. Class-specific differential detection in diffractive optical neural networks improves inference accuracy (2019)

DOI. 10.1117/1.AP.1.4.046001

Innovation. A basic limitation of many all-optical classifiers is that the output detector measures *nonnegative intensity*, whereas electronic classifiers naturally compare signed logits. This paper introduces a very specific fix: **class-specific differential detection**. Each class gets a *positive* and a *negative* detector, and the class score is the difference between the two. That is a *new method* rather than a new application. It makes D^2NN inference more "classifier-like" and substantially boosts accuracy because the optical system can now learn signed evidence for and against each class instead of only putting more light on one detector than another.

Experiment / system setup. Status: numerical study in the accessible sources. The paper analyzes a **5-layer diffractive classifier** with on the order of 2×10^5 trainable neurons and **10 positive/negative detector pairs** for ten-class tasks. I did not recover a physical hardware section for this paper from the accessible primary text during this session, so I treat it as a **simulation-focused design paper**. The practical takeaway is architectural: if one later fabricates such a system, the readout plane should be designed as **paired detectors per class**, not a single detector per class.

4. Analysis of diffractive optical neural networks and their integration with electronic neural networks (2020)

DOI. 10.1109/JSTQE.2019.2921376

Innovation. This paper is important because it turned D^2NNs from a striking prototype into an engineering design space. It asks: what happens if layers are closer together, if modulation is phase-only versus complex-valued, if the loss is changed, and if a small electronic network is appended after the optics? The main innovation is the **hybrid opto-electronic paradigm**: use the diffractive network as an ultra-low-power optical front-end that *compresses* or partially solves the task, then let a lightweight digital back-end finish the decision. That is a specific, practically useful design rule for real systems.

Experiment / system setup. Status: numerical study. The accessible full text reports **5 diffractive layers** throughout the analysis. Two connectivity regimes were compared: a more fully connected design with $\Delta z = 40\lambda$ spacing and a more compact partially connected design with $\Delta z = 4\lambda$. The paper reports representative blind-test accuracies of **97.18%** (phase-only) and **97.81%** (complex-valued) on MNIST, and shows a jointly optimized hybrid system reaching **98.71%** while compressing the optical output to a 10×10 **detector array**. Detector-array options of 10×10, 25×25, and 50×50 were analyzed. No fabricated optical hardware is described in the accessible primary text for this paper.

5. Misalignment resilient diffractive optical networks (2020)

DOI. 10.1515/nanoph-2020-0291

Innovation. This paper addresses one of the most practical weaknesses of fabricated D^2NNs: the trained design assumes perfect axial and lateral alignment, but real layers shift. The innovation is a training strategy the authors describe as **vaccination**: during optimization, each layer is randomly perturbed in 3D so the network learns to be insensitive to assembly and fabrication errors. This is a *new method* for making passive optical networks manufacturable. The point is not just better simulation robustness; it is a direct recipe for building D^2NNs that still work after printing, mounting, and re-assembly.

Experiment / system setup. Status: physical THz demonstration, and the supplement adds most of the key hardware numbers. The experimental device consisted of the **input object plane, 5 diffractive layers, and a 10-aperture output plane**. The active modulation area of each diffractive layer was 8×8 **cm**, while the total printed layer size was 12.8×12.8 **cm** with a surrounding slab thickness of **0.9 mm**. Fabrication used **Objet30 Pro** and the diffractive material parameters at **400 GHz** were reported as $n = 1.7227$ and $\kappa = 0.031$. The lateral sampling interval / minimum feature size was **0.4 mm** (about 0.53λ). Each feature thickness was modeled as a **0.5 mm** base plus up to **1 mm** trainable relief, giving a physical thickness range of **0.5–1.5 mm**; the printed design used **16 quantization levels** with **62.5 μm** spacing. The output plane contained **10 apertures**, each of size 4.8×4.8 **mm**. The THz source used a **VDI WR2.2 AMC** driven at **11.111 GHz** and multiplied by $36\times$ to generate continuous-wave **0.4 THz** illumination; the horn was positioned about **60 cm** from the input plane. Readout used a **VDI single-pixel mixer/AMC detector** with a local oscillator at about **11.083 GHz**, scanned on **two Thorlabs NRT100** stages with a **1 mm** step size. The down-converted signal chain included **Mini-Circuits ZRL-1150-LN+** amplification, a **KL Electronics 3C40-1000/T10-O/O** bandpass filter, an **HP 8495B** attenuator, a **Mini-Circuits ZX47-60** power detector, and a **Stanford Research SR830** lock-in amplifier. The supplement also explains the mechanical misalignment test bed: the holder let the **third layer** be moved among **13 discrete positions**. This SI readout makes the paper much more concrete as an engineering study of **tolerance-aware, five-layer printed THz D^2NN hardware** rather than a generic robustness claim.

6. Scale-, shift-, and rotation-invariant diffractive optical networks (2021)

DOI. 10.1021/acsphotonics.0c01583

Innovation. The main idea here is to build invariance into the optical processor itself rather than relying on digital preprocessing or postprocessing. The network is trained so that the optical transformation remains useful under **scale changes, translations, and rotations** of the input. This is a *new method* because it pushes geometric robustness into the passive diffractive stack. For someone new to the paper: the goal is not a better classifier in general, but a classifier whose optical field distribution remains class-separable even when the object is moved, resized, or rotated before the light reaches the first layer.

Experiment / system setup. Status: numerical study in the accessible sources. The paper uses data augmentation and invariance-aware optimization of diffractive layers, but I did not recover a primary-source hardware section with printer, material, or real-device parameters during this session. Therefore the paper is best treated as an **optical-learning design paper** showing how

invariance can be baked into a D^2NN objective function.

7. All-optical information-processing capacity of diffractive surfaces (2021)

DOI. 10.1038/s41377-020-00439-9

Innovation. This paper is a capacity paper rather than an application paper. Its main contribution is to ask, in a mathematically explicit way, *how much linear information processing a set of passive diffractive surfaces can implement*. That is a *new paradigm-level clarification*: instead of only reporting that D^2NNs work on some dataset, the paper studies the linear transformation space accessible to stacked diffractive surfaces and how that space scales with the number of surfaces and degrees of freedom. This is important for understanding what D^2NNs can and cannot represent before one even picks a task.

Experiment / system setup. Status: theory/numerical paper. The accessible sources do not present a physical printer/SLM implementation. The paper should be read as a **representation-capacity analysis** for passive diffractive processors, useful for deciding how many surfaces and spatial degrees of freedom are needed for later designs.

8. Ensemble learning of diffractive optical networks (2021)

DOI. 10.1038/s41377-020-00446-w

Innovation. A single D^2NN can be brittle, especially on harder datasets. This paper introduces **ensemble learning** into diffractive optics: instead of fabricating one network, train multiple D^2NNs and combine their predictions. The contribution is not simply "more networks improve accuracy"; the paper shows how independent optical solutions can be selected and combined so that their errors decorrelate. That is a *new method* for turning passive optical inference from a single-solution design problem into a committee-of-experts design problem.

Experiment / system setup. Status: numerical study in the accessible sources. The paper focuses on the architecture and selection/combination strategy of multiple diffractive networks rather than a new fabrication platform. I did not recover a primary-source real-device section in this session, so the hardware takeaway is conceptual: if the task is difficult, **multiple separately trained diffractive processors** may be a better deployment unit than a single one.

9. Spectrally encoded single-pixel machine vision using diffractive networks (2021)

DOI. 10.1126/sciadv.abd7690

Innovation. This paper is one of the clearest examples of D^2NNs doing something that is not a standard camera. The innovation is to encode task-relevant spatial information into the **spectrum** measured at a **single output pixel**. In one mode, the spectral response itself is enough for classification; in another, the spectrum is passed to a lightweight electronic decoder to reconstruct the object. So the paper introduces a *new application*—single-pixel machine vision—and also a *new method*: use wavelength as a coding dimension for machine perception instead of forming a conventional image.

Experiment / system setup. Status: physical broadband THz demonstration. The experimental platform used an **input aperture, an input object, 3 transmissive diffractive layers, and an output aperture**. The layers were fabricated with **Objet30 Pro** (**Stratasys**) using **VeroBlackPlus RGD875**. The input aperture was 1×1 **cm**, and a 2×2 **mm** output aperture was placed right before the detector. The THz-TDS system used a **mode-locked Ti:sapphire laser** (**Coherent Mira-HP**) at a center wavelength of **780 nm**; both the THz source and detector were **plasmonic photoconductive nanoantenna arrays**. The detected signal was amplified by a **Femto DHPCA-100** current preamplifier and read with a **Zurich Instruments MFLI** lock-in amplifier. For each measurement, **10 time-domain traces** were captured over **5 s** and averaged; the temporal span was **400 ps**, with reported **SNR** > 90 **dB** and observable bandwidth > 5 **THz**. The detector included an integrated **Si hemispherical lens**, which the forward model approximated as a flat Si slab of thickness **0.5 cm** and refractive index **3.4**. The supplementary material provides the layer-design details: the minimum printed feature size was **0.5 mm**; the simulation grid was oversampled to **0.25 mm**; each diffractive feature thickness was written as $h = q(h_m) + h_{\text{base}}$ with $h_m \in [0, 0.75]$ **mm**, $h_{\text{base}} = 0.2$ **mm**, and a **12-level quantizer** with **62.5** μ**m** steps. The input/output apertures and 3D-printed objects were coated with aluminum. For the standard MNIST classifier configuration, the **10 class wavelengths** were selected between **1.00 mm** and **1.45 mm** with **0.05 mm** spacing. So this is not merely a "single-pixel idea" in the abstract; it is a concretely realized **three-layer printed broadband THz spectral encoder for machine vision**.

10. Terahertz pulse shaping using diffractive surfaces (2021)

DOI. 10.1038/s41467-020-20268-z

Innovation. This paper shifts D^2NNs from spatial-pattern recognition to **temporal waveform engineering**. The new idea is that passive diffractive surfaces can be trained so that a broadband THz pulse emerges with a desired temporal shape or pulse width. The paper also introduces a very useful concept akin to **physical transfer learning**: some layers can be kept fixed while only later layers are retrained for a new target pulse response. So the innovation is both a *new application* (pulse shaping) and a *new method* for reusing previously optimized optical hardware.

Experiment / system setup. Status: physical broadband THz demonstration, and the methods/supplement provide a well specified hardware description. Each pulse-shaping network consisted of **4 trained diffractive layers**, with **3 cm** axial spacing between successive layers. The input aperture was a square of edge length **0.8 cm**; the output aperture was **0.2 cm**; the output aperture was placed **10 cm** after the last diffractive layer. The diffractive layers and apertures were fabricated with **Objet30 Pro** (**Stratasys**); the apertures were aluminum coated, and the layers were mounted in a 3D-printed alignment holder. The THz-TDS setup used a **Ti:sapphire laser** (**Coherent Mira HP**) to generate femtosecond optical pulses, a **plasmonic photoconductive THz source**, a **plasmonic photoconductive THz detector**, and a **Newport IMS300LM** optical delay line. The detected signal was amplified by a **Femto DHPCA-100** transimpedance preamplifier and read by a **Zurich Instruments MFLI** lock-in amplifier. The time sampling resolution was **12.5 fs** over a **400 ps** window; **10 traces** were averaged per measurement; the reported performance was about **90 dB SNR** over a **5 THz** noise-equivalent bandwidth. The forward model sampled **300 discrete frequencies** between **3 GHz** and **1 THz**. The minimum lateral feature size was **0.5 mm**; each feature was oversampled by $2\times$ in simulation; the physical thickness of each feature was decomposed into a **0.1 mm base height** and a trainable modulation

height between **0 and 1 mm**. This makes the paper a clearly specified **four-layer, 3D-printed broadband THz pulse shaper**, not just a general argument that diffractive networks might shape pulses.

11. Computer-free, all-optical reconstruction of holograms using diffractive networks (2021)

DOI. 10.1021/acsphotonics.1c01365

Innovation. Here the D^2NN is trained to do **hologram reconstruction itself**, including suppression of the usual twin-image artifact of in-line holography. The contribution is conceptually important because hologram reconstruction is normally seen as a computational imaging problem solved after measurement. This paper asks the opposite question: can the optical hardware itself perform that inversion? That is a *new application* of diffractive computing to holography and a concrete example of replacing digital post-processing with passive learned optics.

Experiment / system setup. Status: conceptual / numerical in the accessible primary text. The accessible sources indicate a design with **5 transmissive diffractive layers** spanning roughly **225 wavelengths** axially, but I did not recover enough primary-source fabrication detail in this session to call it a fully documented hardware paper. I therefore treat it as a **numerically demonstrated diffractive hologram-reconstruction framework** with clear layer-count information but insufficient recovered fabrication specifics.

12. Polarization multiplexed diffractive computing: all-optical implementation of a group of linear transformations through a polarization-encoded diffractive network (2022)

DOI. 10.1038/s41377-022-00849-x

Innovation. The central insight is that one diffractive stack need not implement only one function. By changing the **input/output polarization channel**, the same physical structure can perform **different linear transformations**. This is a *new method* for multiplexing functions into one passive optical processor. It matters because it moves D^2NNs toward re-use and multifunctionality: instead of fabricating one network per task, one can store a *family* of operators in a single polarization-aware diffractive design.

Experiment / system setup. Status: numerical/theory paper, but the accessible manuscript+SI text gives a much more concrete architecture than a one-line summary would suggest. For the **2-channel** design ($N_p = 2$), the processor used **4 trainable isotropic diffractive layers** plus **one non-trainable linear-polarizer array** inserted between the **2nd** and **3rd** diffractive layers. That polarizer array used four polarization directions: $0°$**,** $45°$**,** $90°$**, and** $135°$. For the **4-channel** design ($N_p = 4$), the network depth was increased to **8 trainable diffractive layers**, and **two** such polarizer arrays were inserted after the **3rd** and **5th** layers. The accessible SI text further states that each polarizer array contained $6 \times 6 = 36$ linear polarizer units, that a polarizer-unit period of about $\sim 4\lambda$ was empirically near-optimal, and that placing two polarizer arrays about $\sim 8\lambda$ apart improved the 4-channel designs. The same SI analysis also warns that adding too many polarizer arrays (e.g., > 6) can significantly hurt transformation accuracy unless more diffractive layers are added. So although this paper does *not* report a fabricated device in the accessible text, the SI does specify a **deep polarization-seeded diffractive architecture** in much more detail

than the previous summary conveyed.

13. All-optical phase recovery: diffractive computing for quantitative phase imaging (2022)

DOI. 10.1002/adom.202200281

Innovation. This paper applies diffractive computing to **quantitative phase imaging (QPI)**. Instead of recording intensity and then running a digital phase-retrieval algorithm, the optical network is trained so the output intensity distribution directly reveals the phase object. That is a *new application* and also a *new paradigm* for QPI: the inverse problem is moved into the design of the passive optical front-end. For a reader new to the work, the main claim is that the diffractive stack itself can approximate the phase-recovery map.

Experiment / system setup. Status: the accessible sources clearly establish the QPI objective, but I did not recover enough primary-source hardware detail in this session to confidently report printer/material/feature-size values. I therefore classify this paper as a **method paper with limited recovered setup detail** from the accessible sources, and I explicitly avoid guessing the missing fabrication parameters.

14. Classification and reconstruction of spatially overlapping phase images using diffractive optical networks (2022)

DOI. 10.1038/s41598-022-12020-y

Innovation. This work tackles a harder inverse problem than standard single-object recognition: **two phase objects overlap spatially**, yet the diffractive network is trained to both classify and reconstruct them. The innovation is task formulation: the D^2NN is not simply a classifier or an imager, but a **joint separator + recognizer + reconstructor** for mixed phase signals. This is a useful example of a diffractive network solving a structured source-separation problem optically.

Experiment / system setup. Status: numerical study in the accessible sources. I did not recover a physical hardware section during this session. The key setup point is therefore algorithmic: the network is trained on **superposed phase objects** and optimized so the output plane contains enough information for both class inference and reconstruction of the mixed inputs.

15. To image, or not to image: class-specific diffractive cameras with all-optical erasure of undesired objects (2022)

DOI. 10.1186/s43593-022-00021-3

Innovation. This is one of the most original application papers in the D^2NN literature. The idea is to design a camera that **forms an image only for desired classes** and **optically erases everything else**. So the output sensor never sees the suppressed objects in the first place. That is both a *new application* (privacy-/policy-aware optical sensing) and a *new paradigm*: the optical system performs a semantic operation before any conventional image is created. The paper also explores class-specific transformation/permutation/encryption variants, showing that imaging itself can be made class-conditional.

Experiment / system setup. Status: both numerical design and physical THz demonstration. In the numerical monochromatic designs, the paper reports $\lambda = 0.75$ **mm**, 120×120 **diffractive features per layer**, **0.4 mm** feature size, and about **20 mm** axial distance between successive planes. For the physical THz experiments, the accessible sources indicate printed layers fabricated with **PR110** on a **CADworks3D** printer, while holders and test objects were made with **Objet30 Pro**; some test objects were coated with aluminum foil. Illumination used a **0.4 THz CW** VDI multiplier-chain source, and the output was scanned using a VDI detector on motorized **Thorlabs NRT100** stages. The paper also reports vaccination-style tolerances of roughly **0.4 mm lateral** and **1.5 mm axial** perturbation in one of the validated designs.

16. All-optical image classification through unknown random diffusers using a single-pixel diffractive network (2023)

DOI. 10.1038/s41377-023-01116-3

Innovation. The problem here is unusually hard: classify an object when it is hidden behind an **unknown random diffuser**, without first reconstructing a clean image. The paper's answer is to combine a diffractive processor with **single-pixel spectral readout**. That is a *new application*—classification through unknown scattering—and also a *new method*: use wavelength-dependent single-pixel statistics as the task representation instead of trying to invert the diffuser. The important conceptual point is that the optical network learns a robust discriminative measurement, not a restoration pipeline.

Experiment / system setup. Status: physical THz demonstration. The accessible sources describe a **single 3D-printed diffractive layer** with about 120×120 **neurons** and **0.4 mm** neuron size. The experiment used a broadband THz-TDS system and compared the output at two wavelengths, reported as approximately $\lambda_0 = 0.9$ **mm** and $\lambda_1 = 1.2$ **mm**. The physical device was used as a **single-layer single-pixel classifier** through previously unseen random diffusers, which is precisely why the paper is interesting: the optical hardware is tiny, yet the task is extremely nontrivial.

17. Data class-specific all-optical transformations and encryption (2023)

DOI. 10.1002/adma.202212091

Innovation. This paper generalizes class-specific imaging into class-specific **linear optical operators**. Different classes can be assigned different transforms, including identity-like recovery for one class and suppression or alternative transforms for others. It also shows **all-optical encryption**: only the correct optical "key" or inverse transform retrieves the intended information. This is a *new method* because the novelty is not simply encrypting an image, but making the optical transformation itself depend on **data class**. That lets one think of D^2NNs as optical access-control devices, not only cameras or classifiers.

Experiment / system setup. Status: physical near-infrared / telecom-band demonstration. The accessible sources report an experimental **2-layer transmissive D^2NN** operating at **1550 nm**. Fabrication used **two-photon polymerization**; input objects were made by photolithography and outputs were captured by a CCD. I did not recover the exact printer model, resist name, or voxel/pixel size from the accessible primary text in this session, so I state only the recovered facts: **two physical layers**, **1550 nm operation**, and **two-photon-polymerized diffractive**

hardware.

18. Universal linear intensity transformations using spatially incoherent diffractive processors (2023)

DOI. 10.1038/s41377-023-01234-y

Innovation. Earlier D^2NN papers usually assume coherent illumination and complex field control. This work asks a more practical question: what can be done under **spatially incoherent monochromatic light**, where one only manipulates intensity transformations? The paper introduces a framework for **universal linear intensity transforms** in that regime. This is a *new paradigm* because it pushes diffractive computing closer to natural-light / conventional-illumination settings instead of ideal coherent lab conditions.

Experiment / system setup. Status: theory/numerical in the accessible sources. The abstract gives the main operator-scaling rule and universal-approximation condition, but I did not recover a physical implementation section. The key setup takeaway is conceptual: the design target is a **linear intensity operator under incoherent illumination**, not a coherent complex-field map.

19. Massively parallel universal linear transformations using a wavelength-multiplexed diffractive optical network (2023)

DOI. 10.1117/1.AP.5.1.016003

Innovation. This paper shows how to make one diffractive processor implement **many different arbitrary linear transforms**, one per wavelength channel. The core innovation is **wavelength multiplexing of universal linear operators**. That is more specific than just "multicolor optics": each wavelength is a different computational instruction set. This creates a route to massively parallel optical linear algebra where a single broadband component behaves like many distinct processors.

Experiment / system setup. Status: both numerical and **physical THz proof-of-concept**; this is one of the entries that changed the most after reading the combined manuscript+SI PDF. The experimental validation used a wavelength-multiplexed permutation task with $N_w = 2$ wavelength channels and $N_i = N_o = 32$. The two experimental wavelengths were $\lambda_1 = 0.667$ **mm** (**450 GHz**) and $\lambda_2 = 0.698$ **mm** (**430 GHz**), each assigned to a different target permutation matrix. The processor itself used **3 diffractive layers**, each containing 120×120 features with a lateral size of **0.4 mm** (about $0.59\lambda_m$); the axial spacing between adjacent planes, including the input/output planes, was **20 mm** (about $29.3\lambda_m$). Training used **55,000** randomly generated input-output field pairs. The diffractive layers were fabricated using **PR110** on a **CADworks3D** printer, while the input objects and holders were 3D printed using **Objet30 Pro** and the input objects were coated with aluminum foil. The experimental THz bench used a **VDI WR2.2 AMC** followed by a horn antenna; RF inputs of **11.944 GHz** and **12.500 GHz**, multiplied by 36×, produced the two illumination frequencies. The diffractive network was positioned **600 mm** from the source aperture. Output scanning used a **VDI single-pixel mixer/AMC detector** mounted on **two Thorlabs NRT100** stages with a **2 mm** step size. The local oscillator frequencies were **11.917 GHz** and **12.472 GHz** for the two channels, and the readout chain included **Mini-Circuits ZRL-1150-LN+**, **KL Electronics 3C40-1000/T10-O/O**, **HP 8495B**, **Mini-Circuits ZX47-60**, and **Stanford Research SR830**. So, after the SI/main-PDF read, this entry should *not* be

labeled numerical-only: it includes a concrete **three-layer, wavelength-multiplexed printed THz experimental demonstration**.

20. Snapshot multispectral imaging using a diffractive optical network (2023)

DOI. 10.1038/s41377-023-01135-0

Innovation. The paper replaces a conventional spectral filter array with a **learned virtual spectral filter array** created by a diffractive network. In plain language: instead of depositing many different spectral filters on the detector, the optics itself learns to split wavelength information across the output field so that a snapshot encodes a multispectral cube. This is a *new application* (snapshot multispectral imaging) and a *new method* for avoiding bulky or expensive dispersive/filter hardware.

Experiment / system setup. Status: physical THz demonstration plus visible-spectrum numerical design. The visible numerical design used **5 diffractive layers** with roughly **0.76 million trainable features**. The experimental THz imager used **3 printed diffractive surfaces**, each with 100×100 **features**, **0.5 mm** minimum feature size, and about 5×5 **cm** active area. The layers were fabricated with **Objet30 Pro**. The reported experimental wavelengths were approximately **0.8, 0.75, 0.70, and 0.66 mm** (roughly **0.375–0.45 THz**). This is a real **three-layer printed THz snapshot multispectral imager**.

21. Quantitative phase imaging (QPI) through random diffusers using a diffractive optical network (2023)

DOI. 10.37188/lam.2023.017

Innovation. This paper combines two hard things at once: **phase imaging** and **unknown scattering**. The D^2NN is trained so that phase objects hidden behind random diffusers can still be quantitatively reconstructed. That is a *new application* because it targets all-optical QPI through disorder. Compared with standard QPI, the conceptual leap is that the optical processor is asked to be both a phase-retrieval operator and a diffuser-robust inverse mapper.

Experiment / system setup. Status: method paper with limited recovered setup detail from the accessible sources. The abstract reports an axial extent of roughly **70 wavelengths** and discusses compatibility with CCD/CMOS imaging, but I did not recover the full experimental hardware parameters in this session. I therefore keep the setup description conservative rather than guessing unverified fabrication values.

22. Learning diffractive optical communication around arbitrary opaque occlusions (2023)

DOI. 10.1038/s41467-023-42556-0

Innovation. The novel problem here is communication **around** an opaque blocker rather than through a clear line of sight. The paper jointly optimizes an encoder and a passive diffractive decoder so the message can bypass arbitrary occlusions and still be recovered. A particularly useful idea is that when the occlusion changes, one may only need to **retrain the electronic/encoding**

side rather than redesigning the passive optical decoder. That makes this paper a *new application* of D^2NNs to optical communication and a *new system method* for adapting around blockers.

Experiment / system setup. Status: physical THz demonstration. The accessible sources report a **single-layer diffractive decoder** fabricated with **Objet30 Pro**; the phase-encoded message masks had a **0.2 mm** base thickness, and the opaque occlusion was made by coating a 3D-printed substrate with aluminum. Operation used $\lambda = 0.75$ **mm** and a **continuous-wave 0.4 THz** VDI source; the receiving side used a VDI single-pixel detector with a local oscillator near **11.0833 GHz**. The detector aperture was approximately 0.5×0.1 **mm**, and output scanning used roughly **2 mm** steps. So the validated device is a **one-layer printed THz decoder for non-line-of-sight / occlusion-aware communication**.

23. Rapid sensing of hidden objects and defects using a single-pixel diffractive terahertz sensor (2023)

DOI. 10.1038/s41467-023-42554-2

Innovation. This paper is a strong example of D^2NNs replacing imaging with **task-specific sensing**. Instead of forming an image of hidden defects and then analyzing it, the system directly maps the scene to a **single-pixel spectral signature** that reveals whether and where hidden defects exist. The novelty is very concrete: a **non-imaging THz defect sensor** learned end-to-end for hidden-object/defect interrogation, including defects embedded in silicon. This is a *new application* and also an important sensor-design philosophy for diffractive computing.

Experiment / system setup. Status: physical broadband THz demonstration. The sensor used **4 diffractive layers total**: **2 front-end layers before the sample** and **2 back-end layers after the sample**. Each layer had about 100×100 **features** with **0.5 mm** lateral feature size. The design wavelengths were approximately $\lambda_1 = 0.8$ **mm** and $\lambda_2 = 1.1$ **mm**. Reported thickness parameters include **0.6 mm** base thickness and **0.8 mm** maximum trainable relief. The diffractive layers were printed with a **Formlabs Form 3**; the holder was made by **Objet30 Pro**. The measurement platform was a broadband **THz-TDS** system pumped by a **780 nm Ti:Sapphire** laser, with THz pulses peaking around **500 GHz** and bandwidth extending to several THz.

24. Unidirectional imaging using deep learning-designed materials (2023)

DOI. 10.1126/sciadv.adg1505

Innovation. This paper explores a very unusual optical function: **the material images in one direction but not in the reverse direction**. Because ordinary reciprocal linear optics usually treats forward and backward propagation symmetrically, the important conceptual contribution is to show that learned diffractive structuring can create **direction-dependent functionality** even with passive linear materials. This is a *new paradigm* for D^2NN-based materials and a direct extension of class-specific imaging into one-way imaging behavior.

Experiment / system setup. Status: physical THz demonstration. The experimentally validated design used **3 diffractive layers**, each with 100×100 **learnable features**. The feature size was about **0.48 mm** (reported as roughly **0.64**λ), with axial spacing around **26.7**λ. The field of view on each side was about 36×36 **mm**, sampled into 15×15 pixels of **2.4 mm**. The layers were fabricated from **PR110** using **CADworks3D**; holders and test objects were made with **Objet30**

Pro, and some objects were aluminum coated. Illumination used a **400 GHz CW** VDI source generated from **11.1111 GHz**×36; the horn was about **600 mm** away, and output scanning used a VDI detector on **Thorlabs NRT100** stages with **1 mm** step size. Vaccination tolerances of about **0.48 mm lateral** and **1.5 mm axial** were also reported.

25. Complex-valued universal linear transformations and image encryption using spatially incoherent diffractive networks (2024)

DOI. 10.1117/1.APN.3.1.016010

Innovation. This paper extends the 2023 incoherent-light framework from nonnegative intensity transforms to **arbitrary complex-valued linear transformations** and associated **image-encryption** tasks under spatially incoherent illumination. The conceptual advance is important: it shows that incoherent-light diffractive processors are not restricted to very limited intensity remappings but can approximate much richer operator families. That is a *new paradigm-level extension* of diffractive computing beyond coherent optical computing.

Experiment / system setup. Status: theory/numerical in the accessible sources. I did not recover a physical hardware section with SLM or fabricated diffractive layers during this session. The paper's value is primarily in showing **operator universality and encryption under incoherent light**, which broadens the feasible illumination conditions for diffractive computing.

26. Information-hiding cameras: optical concealment of object information into ordinary images (2024)

DOI. 10.1126/sciadv.adn9420

Innovation. This is a very distinctive paper. The idea is not to erase unwanted content, but to **hide true scene information inside an ordinary-looking image**. The optical front-end is trained so that the camera output resembles a benign dummy image (for example, an image from a specific class), while a learned electronic decoder can recover the real object. This is a *new application*—optical concealment / steganographic imaging—and also a *new system method*: a jointly optimized diffractive encoder plus digital decoder that intentionally makes the sensor output misleading to an uninformed observer.

Experiment / system setup. Status: both numerical design and physical THz demonstration. For the **monochromatic numerical model**, the information-hiding diffractive camera used **5 transmissive phase-modulation layers** that axially span $< 110\lambda$ in total; each layer contained 120×120 trainable features with a lateral size of about 0.53λ (**0.4 mm** at $\lambda = 0.75$ **mm**), and the spacing between successive planes was 26.7λ. For the **multispectral visible design**, the system again used **5 layers**, with wavelengths **630/530/450 nm**, 120×120 features per layer, feature size **200 nm**, and about **20** μ**m** spacing between successive planes. For the **THz proof-of-concept experiment**, the authors first trained a **single-layer** information-hiding diffractive processor at $\lambda = 0.75$ **mm**; that layer had 100×100 trainable features, each of lateral size about 0.67λ (**0.5 mm**), with the axial distance between the input/output planes and the diffractive layer set to 26.7λ (**20 mm**). After training, the diffractive layer, test objects, and holder were fabricated using **Stratasys Objet30 Pro**, and the test objects were coated with aluminum foil. The physical diffractive camera was then tested using a **THz continuous-wave scanning system** at $\lambda = 0.75$ **mm**; a digital 2×2 **binning** step was applied after the measurement to match the simulated

object-feature sampling. The paper also reports a **two-layer THz experimental variant**, with the same **20 mm** inter-plane spacing and a vaccinated CNN decoder used to improve robustness against physical misalignment. In this case, the supplementary material that I could access mainly listed **Figs. S1–S3** and movie legends, so the important setup compensation here comes mostly from the main paper and methods text rather than from SI-only hardware tables.

27. All-optical complex field imaging using diffractive processors (2024)

DOI. 10.1038/s41377-024-01482-6

Innovation. This paper is about recovering the **full complex optical field**—both amplitude and phase—using only intensity measurements after a learned diffractive processor. The key idea is to train the output field of the processor so that separate spatial regions encode amplitude and phase information. For someone new to the paper: instead of recording an interferogram and running digital phase retrieval, the passive optics directly generate an intensity pattern from which the complex field is read out. That is a *new application* of D^2NNs to complex-field metrology and a strong example of replacing a computational inverse step with optical design.

Experiment / system setup. Status: both numerical design and physical THz demonstration. One numerical design used **5 layers** with about 200×200 **diffractive features per layer** at roughly $\lambda/2$ lateral pitch. The physical THz experiment used **3 layers**, each with 120×120 **features** of lateral size about $0.516\lambda_m$; adjacent layers were separated by **20 mm** (about $25.8\lambda_m$), for a total axial depth of about $103.2\lambda_m$. The printed structures were fabricated using **PR110** on **CADworks3D**; the reported wavelengths were about $\lambda_1 = 0.75$ **mm** and $\lambda_2 = 0.8$ **mm**. Thus the paper includes a real **three-layer printed THz complex-field imager**.

28. All-optical phase conjugation using diffractive wavefront processing (2024)

DOI. 10.1038/s41467-024-49304-y

Innovation. Optical phase conjugation is usually associated with nonlinear optics or specialized phase-conjugate mirrors. This paper asks whether a **passive learned diffractive stack** can approximate that operation. The answer is yes: the diffractive processor is trained to produce a phase-conjugated version of the incoming wavefront, and the paper also discusses a mirror-assisted phase-conjugate arrangement. The innovation is therefore a *new application* of D^2NNs to wavefront reversal and scattering control, not merely classification or imaging.

Experiment / system setup. Status: physical THz demonstration. The accessible sources indicate an experimental **3-layer 3D-printed diffractive processor** operating in the THz regime and spanning only a few tens of wavelengths axially. Publicly accessible snippets suggest the printed layers and test objects were made using the same 3D-printing workflow as related Ozcan-group THz papers (including **Objet30 Pro**), but I did not recover the full primary fabrication table in this session. The most reliable recovered hardware facts are therefore: **three physical layers**, **THz operation**, and a **3D-printed passive phase-conjugation processor**.

29. Nonlinear encoding in diffractive information processing using linear materials (2024)

DOI. 10.1038/s41377-024-01529-8

Innovation. This paper deals with a subtle but important question: can one obtain useful **nonlinear behavior** from an optical processor built only from **linear materials**? The answer is that the *encoding strategy* matters. By changing how information is represented at the input and read at the output, the overall system can implement useful nonlinear maps even though the propagation physics remains linear. This is a *new paradigm clarification*. It does not claim that passive linear optics suddenly becomes a digital deep network, but it gives a careful account of what kinds of nonlinearity can be created through encoding/decoding choices.

Experiment / system setup. Status: conceptual/theory and numerical paper in the accessible sources. I did not recover a dedicated physical printer/SLM implementation during this session. The practical value of the paper is to guide how later D^2NN experiments should choose phase/amplitude encoding and replication schemes if they want stronger effective nonlinearity from linear optical hardware.

30. All-optical image denoising using a diffractive visual processor (2024)

DOI. 10.1038/s41377-024-01385-6

Innovation. This paper turns denoising into an all-optical front-end operation. Instead of digitally filtering noise after capture, the diffractive processor is trained so that **signal energy is directed into the output field of view while noise is rejected outside it**. That is a *new application* and a useful design principle: D^2NNs can act as **analog visual preprocessors** that clean the image before any electronic inference. The paper is especially interesting because it handles not just one synthetic noise case but also rendering/propagation distortions relevant to optical systems.

Experiment / system setup. Status: physical THz demonstration. The experiment used a **3-layer diffractive denoiser**. The operating wavelength was about 0.75 **mm** (**0.4 THz**). The input and output field of view were each about $40\lambda \times 40\lambda$ (roughly 3×3 **cm**), while the diffractive layers were about $66.67\lambda \times 66.67\lambda$ (roughly 5×5 **cm**). The feature width was about 0.667λ; material refractive index was reported as approximately $1.6518 + j0.0612$; layer thickness values ranged from about **0.5 mm** to **1.65 mm**. Fabrication used **Objet30 Pro**. Illumination came from a **0.4 THz CW** source built from a WR9.0/WR4.3/WR2.2 multiplier chain; the detector pinhole was about 0.25×0.5 **mm** with **0.75 mm** scan step size.

31. Pyramid diffractive optical networks for unidirectional image magnification and demagnification (2024)

DOI. 10.1038/s41377-024-01543-w

Innovation. The novel architectural idea is to use a **pyramid / tapered geometry** rather than layers of equal size. When the optical task is one-way magnification or demagnification, a tapered stack can better match the directional flow of spatial information and reduce unnecessary degrees of freedom. This is a *new method* for unidirectional imaging tasks. In plain language, the optics are shaped like the transformation they are supposed to implement.

Experiment / system setup. Status: physical THz validation is reported in the accessible sources, including magnification factors such as $M = 2$ and $M = 3$. However, I did not recover a complete primary-source fabrication table during this session, so I cannot confidently state the exact printer, material, or feature pitch without over-claiming. I therefore retain the main setup fact—**experimentally validated THz pyramid D^{2}NNs for one-way magnification/demagnification**—and explicitly leave finer fabrication values unspecified.

32. Multiplexed all-optical permutation operations using a reconfigurable diffractive optical network (2024)

DOI. 10.1002/lpor.202400238

Innovation. This paper shows how to make a D^{2}NN **reconfigurable** without an SLM. Each layer can be physically rotated among **four orientations** (0, 90, 180, 270 degrees), and the combination of rotations across layers selects the implemented optical permutation. With K **layers**, the number of available permutations scales as 4^K. That is a highly specific *new method*: using discrete mechanical configuration states to multiplex many all-optical operators into one printed diffractive system. It is a clever bridge between completely static hardware and fully programmable modulators.

Experiment / system setup. Status: physical THz demonstration. The accessible sources report a reconfigurable system with **3D-printed diffractive layers** and an experimentally demonstrated space of up to **256 permutations** using $K = 4$ layers/configuration degrees. I did not recover the full primary fabrication table with printer/material/feature-size values in this session, so I avoid inventing them. The critical hardware point is the **discrete mechanical reconfiguration of the diffractive layers** to choose among many optical permutations in a real THz system.

33. Massively parallel and universal approximation of nonlinear functions using diffractive processors (2025)

DOI. 10.1186/s43593-025-00113-w

Innovation. This paper moves beyond classification/imaging to **function computation**. Its main claim is that, with appropriate encoding, a diffractive processor can serve as a **universal approximator of band-limited nonlinear functions**, and can do so **massively in parallel**. The new paradigm is that D^{2}NNs are not just optical classifiers or filters; they can be general analog optical computers for families of nonlinear maps. The paper also emphasizes parallelism: very large numbers of nonlinear functions can be approximated simultaneously at wavelength-scale spatial density.

Experiment / system setup. Status: both numerical and physical optical demonstration. The accessible abstract reports numerical approximation of about **one million nonlinear functions** and an experimental demonstration of **35 nonlinear functions in parallel**. The physical experiment used a compact **SLM + image-sensor** setup. However, in the accessible primary text I recovered, the exact **SLM model**, **native resolution**, and **reflection/transmission type** were not stated clearly enough for me to quote them with confidence. I therefore report the validated facts—**physical SLM-based demonstration** and **35 parallel nonlinear functions**—and explicitly leave the unrecovered SLM part-number unspecified.

34. Model-free optical processors using in situ reinforcement learning with proximal policy optimization (2026)

DOI. 10.1038/s41377-025-02148-7

Innovation. This paper introduces a major training shift: **model-free, in situ reinforcement learning** for optical hardware. Earlier D^2NN work usually assumes a differentiable forward model and backpropagation through simulation. Here the optical system is trained directly using **proximal policy optimization (PPO)** without requiring an accurate forward model. That is a *new training paradigm*. It matters because real optical hardware often drifts, is hard to model perfectly, or includes unknown aberrations; PPO lets the physical system itself become part of the optimization loop.

Experiment / system setup. Status: physical SLM-based demonstration. The accessible primary text describes experiments including focusing through a diffuser, hologram generation, aberration correction, and optical image classification. In one image-classification task, an 800×800 **region of an SLM** served as a **single trainable optical layer**, while the same SLM also provided the input interface; output intensity was recorded by an image sensor/camera. I did not recover the exact **SLM manufacturer/model** or whether the modulator was reflective or transmissive from the accessible text, so I state that uncertainty explicitly. The most reliable recovered setup facts are **single-layer SLM-based in situ training**, **camera-based intensity readout**, and **PPO optimization directly on physical hardware**.

35. Structural vibration monitoring with diffractive optical processors (2026)

DOI. 10.1126/sciadv.aea1712

Innovation. This paper applies diffractive computing to **structural health monitoring**. The optical processor is jointly optimized with a shallow neural back-end so the measured field reveals the **3D vibration spectrum** of a vibrating structure. The innovation is a *new application*: instead of imaging a scene, the diffractive processor is designed for remote dynamic-state estimation. This is exactly the kind of task-specific analog front-end that makes diffractive optics attractive: the optics are trained to extract the physically meaningful signal before the digital stage ever sees the data.

Experiment / system setup. Status: physical millimeter-wave / laboratory structural-monitoring demonstration. The accessible abstract describes a **laboratory-scale building** mounted on a **programmable shake table** and illuminated in the **millimeter-wave** regime, using a **3D-printed diffractive processor** in front of a shallow neural estimator. I did not recover a full primary-source fabrication table with printer model, material, feature size, or exact layer count during this session. Therefore I report the experimentally verified system-level setup—**lab structure + shake table + mm-wave illumination + printed diffractive processor**—and leave the unrecovered fine hardware parameters blank rather than guessing.